GNEST305 · Artificial Intelligence & Data Science

7-Mark Solved Question Bank — Modules 2, 3 and 4 (with full numerical working)

Companion interactive dashboard: GNEST305_7Mark_Dashboard.html (labs, 3-D PCA, knowledge graph, quiz, light/dark theme). Note: pages with blurred text (Module 3 Q4, Q5, Q10 of the first probability page) are solved only where the text was legible.

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1. Spectral decomposition of a 2×2 symmetric matrix [Module 2 · M2 · Q21 · 7 marks]

Given A = [[4,1],[1,3]], perform its spectral decomposition and interpret the results.

Step 1 – Characteristic equation. $\det(A - \lambda I) = (4-\lambda)(3-\lambda) - 1 = \lambda^2 - 7\lambda + 11 = 0$.

Step 2 – Eigenvalues. $\lambda = (7 \pm \sqrt{(49-44)})/2 = (7 \pm \sqrt{5})/2 \rightarrow \mathbf{\lambda_1 = 4.618, \lambda_2 = 2.382}$. (Check: sum = 7 = trace, product = 11 = det.)

Step 3 – Eigenvectors. For λ_1: $(4-4.618)x + y = 0 \rightarrow y = 0.618x \rightarrow v_1 = (0.8507, 0.5257)$. For λ_2: $(4-2.382)x + y = 0 \rightarrow y = -1.618x \rightarrow v_2 = (0.5257, -0.8507)$. Check $v_1 \cdot v_2 = 0.447 - 0.447 = 0$ (orthogonal, as expected for a symmetric matrix).

Step 4 – Decomposition. $A = Q\Lambda Q^T = \lambda_1 v_1 v_1^T + \lambda_2 v_2 v_2^T$, with $Q = [v_1\ v_2]$ and $\Lambda = \mathrm{diag}(4.618, 2.382)$.

Step 5 – Verify. $\lambda_1 v_1 v_1^T = 4.618 \cdot [[0.7236, 0.4472], [0.4472, 0.2764]] = [[3.342, 2.065], [2.065, 1.276]]$; $\lambda_2 v_2 v_2^T = 2.382 \cdot [[0.2764, -0.4472], [-0.4472, 0.7236]] = [[0.658, -1.065], [-1.065, 1.724]]$. Sum = [[4,1],[1,3]] ✓.

Interpretation. A stretches space by 4.618 along v_1 and by 2.382 along v_2 (both positive ⇒ A is positive definite). In PCA, if A were a covariance matrix, v_1 would be the first principal axis carrying 4.618/7 = 66 % of total variance.

Answer: $\lambda_1 = 4.618$ ($v_1 = [0.851, 0.526]$), $\lambda_2 = 2.382$ ($v_2 = [0.526, -0.851]$); $A = \Sigma\, \lambda_i\, v_i v_i^T$

2. Singular Value Decomposition – derivation and 3×3 example

[Module 2 · M2 · Q12, Q13, Q22 · 7 marks]

Derive $A = U\Sigma V^T$ and compute the SVD of A = [[3,1,1],[−1,3,1],[1,1,3]]. How does SVD help in compression?

Derivation. For any m×n matrix A, A^TA is symmetric positive semi-definite ⇒ $A^TA = V\Lambda V^T$ with orthonormal V. Let $\sigma_i = \sqrt{\lambda_i}$ and $u_i = Av_i/\sigma_i$. Then the u_i are orthonormal and $AV = U\Sigma \Rightarrow \mathbf{A = U\Sigma V^T}$.

Step 1 – A^TA. $A^TA = [[11,1,5],[1,11,7],[5,7,11]]$.

Step 2 – Eigenvalues of A^TA (characteristic cubic): $\lambda = 20.094, 10.055, 2.851$ (trace = 33 ✓).

Step 3 – Singular values. $\sigma_1 = \sqrt{20.094} = \mathbf{4.483}$, $\sigma_2 = \sqrt{10.055} = \mathbf{3.171}$, $\sigma_3 = \sqrt{2.851} = \mathbf{1.688}$. Check: $\sigma_1\sigma_2\sigma_3 = 24 = |\det A|$ ✓.

Step 4 – V = eigenvectors of A^TA (columns): $v_1 = (0.441, 0.577, 0.687)$, $v_2 = (0.816, -0.577, -0.039)$, $v_3 = (-0.374, -0.577, 0.726)$ (signs are arbitrary).

Step 5 – U from $u_i = Av_i/\sigma_i$, e.g. $u_1 = A\,v_1 / 4.483 = (2.587, 1.977, 3.079)/4.483 = (0.577, 0.441, 0.687)$.

Compression. Keep the top k singular values: $A \approx \Sigma_{i=1}^{k}\, \sigma_i u_i v_i^T$ (rank-k). Storage falls from m·n to k(m+n+1). Here k=1 keeps $\sigma_1^2/\Sigma\sigma^2 = 20.09/33 = 61$ % of the energy; k=2 keeps 91 %. By Eckart–Young this is the best rank-k approximation.

Answer: σ = 4.483, 3.171, 1.688; $A = U\Sigma V^T$; rank-2 truncation keeps 91 % energy

3. PCA – steps, worked 2-D → 1-D example [Module 2 · M2 · Q15, Q16, Q24 · 7 marks]

Describe the steps of PCA and illustrate how PCA transforms a 2-D dataset into 1-D.

Algorithm. (1) Standardise / mean-centre the data. (2) Compute covariance matrix C. (3) Find eigenvalues & eigenvectors of C. (4) Sort by eigenvalue, keep top k. (5) Project: $Z = X_{centred} \cdot W_k$.

Example data. X = [2,3,4,5], Y = [2.5,3.9,5.2,6.4]. Means: $\bar{x} = 3.5$, $\bar{y} = 4.5$.

Centred: $x' = [-1.5, -0.5, 0.5, 1.5]$; $y' = [-2.0, -0.6, 0.7, 1.9]$.

Covariance (n−1 = 3). Var(x) = 5/3 = 1.667; Var(y) = (4+0.36+0.49+3.61)/3 = 2.820; Cov = (3+0.3+0.35+2.85)/3 = 2.167.

Eigenvalues. $\lambda^2 - 4.487\lambda + (1.667 \cdot 2.820 - 2.167^2) = \lambda^2 - 4.487\lambda + 0.0056 = 0 \rightarrow \mathbf{\lambda_1 \approx 4.485, \lambda_2 \approx 0.001}$.

PC1 direction ∝ (Cov, λ_1−Var(x)) = (2.167, 2.819) → unit vector (0.610, 0.793).

Projection to 1-D. $z = 0.610x' + 0.793y' = [-2.50, -0.78, 0.86, 2.42]$. Explained variance = 4.485/4.487 ≈ **99.98 %**, so one number per point is enough.

Answer: PC1 = (0.610, 0.793) keeps 99.98 % of variance; 2-D data → 1-D scores

4. PCA on Breast-Cancer Wisconsin data (Python) [Module 2 · M2 · Q45 · 7 marks]

Using sklearn's Breast Cancer dataset: apply PCA, compute explained variance ratio, rank and rename the components.

Plan. Load → standardise (PCA is scale-sensitive) → PCA → explained_variance_ratio_ → rank → rename.

```
from sklearn.datasets import load_breast_cancer
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import pandas as pd
X = load_breast_cancer().data
Xs = StandardScaler().fit_transform(X)
pca = PCA().fit(Xs) # (a)
Z = pca.transform(Xs)
evr = pca.explained_variance_ratio_ # (b)
order = evr.argsort()[::-1] # (c) already sorted
names = [f'PC{i+1}_'+('HighVar' if r>0.10 else 'MidVar' if r>0.05 else 'LowVar') for i,r in
enumerate(evr)] # (d)
df = pd.DataFrame(Z, columns=names)
```

Typical result (30 features). PC1 ≈ 44.3 %, PC2 ≈ 19.0 %, PC3 ≈ 9.4 %; cumulative PC1–PC3 ≈ 72.6 %; ≈ 10 components retain 95 %.

Renaming rule. HighVar if ratio > 10 %, MidVar if 5–10 %, LowVar below 5 %. Choose k with np.cumsum(evr) ≥ 0.95 or the scree-plot elbow.

Answer: Standardise → PCA → evr (44.3 %, 19.0 %, 9.4 %…) → rank → rename PCk_High/Mid/LowVar

5. Covariance vs correlation – numerical [Module 3 · M3 · Q1, Q2, Q9 · 7 marks]

Define covariance and differentiate from correlation. Compute covariance for X=[1,2,3], Y=[2,4,5] and Pearson r for X=[10,20,30,40], Y=[15,25,35,45].

Covariance measures how two variables vary together: $\mathrm{Cov}(X,Y) = \Sigma(x_i-\bar{x})(y_i-\bar{y})/(n-1)$. Sign shows direction, but magnitude depends on units. **Correlation** $r = \mathrm{Cov}/(\sigma_x\sigma_Y)$ is its unit-free, standardised form bounded in $[-1, +1]$.

Part 1. $\bar{x} = 2$, $\bar{y} = 11/3 = 3.667$. Deviations: $x' = -1, 0, 1$; $y' = -1.667, 0.333, 1.333$. Products: 1.667, 0, 1.333 → $\Sigma = 3.0$.

Sample covariance = 3.0/2 = **1.5** (population covariance = 3.0/3 = 1.0).

Sample SDs: $\sigma_x = 1$; $\sigma_Y = \sqrt{[(2.778+0.111+1.778)/2]} = \sqrt{2.333} = 1.528 \Rightarrow r = 1.5/(1\times1.528) =$ **0.982**.

Part 2. $\bar{x} = 25$, $\bar{y} = 30$. $x' = -15,-5,5,15$; $y' = -15,-5,5,15$. $\Sigma x'y' = 225+25+25+225 = 500$; $\Sigma x'^2 = \Sigma y'^2 = 500$.

$r = 500/\sqrt{(500\times500)} =$ **+1.00** (perfect positive linear: Y = X + 5).

Answer: Cov = 1.5 (sample), $r_1 = 0.982$, $r_2 = +1$

6. Pearson's coefficient – properties and interpretation [Module 3 · M3 · Q3–Q8 · 7 marks]

Define Karl Pearson's coefficient, list properties, and solve Q5, Q7, Q8.

Definition. $r = \Sigma(x-\bar{x})(y-\bar{y}) / \sqrt{[\Sigma(x-\bar{x})^2 \cdot \Sigma(y-\bar{y})^2]}$. It measures strength and direction of *linear* association.

Properties. (i) $-1 \le r \le +1$. (ii) Independent of change of origin and scale ($r(aX+b, cY+d) = r$ for $ac>0$). (iii) Symmetric: $r_{xy} = r_{yx}$. (iv) Independent variables give $r = 0$, but $r = 0$ does not imply independence (non-linear dependence possible). (v) Geometric mean of the two regression coefficients: $r^2 = b_{yx} \cdot b_{xy}$.

Meaning of +1, −1, 0. +1: perfect increasing line (hours vs. 2×hours). −1: perfect decreasing line (price vs. a fixed budget's units). 0: no linear relation (shoe size vs. IQ).

Q5: X=[2,4,6,8], Y=[10,9,7,5]. $\bar{x}=5$, $\bar{y}=7.75$. $\Sigma x'y' = (-3)(2.25)+(-1)(1.25)+(1)(-0.75)+(3)(-2.75) = -6.75-1.25-0.75-8.25 = -17$. $\Sigma x'^2 = 20$; $\Sigma y'^2 = 5.0625+1.5625+0.5625+7.5625 = 14.75$. $r = -17/\sqrt{(20 \times 14.75)} = -17/17.176 =$ **−0.990** → strong **negative** correlation.

Q7: X=[1..5], Y=[2,4,6,8,10] → Y = 2X exactly, so r = **+1**.

Q8: r = 0.95 → very strong positive linear relationship; $r^2 = 0.9025$, i.e. ≈ 90 % of the variance in one variable is linearly explained by the other. It does **not** prove causation.

Answer: Q5: r = −0.990 (negative); Q7: r = +1; Q8: strong positive, r² ≈ 90 %

7. Regression analysis and the least-squares principle [Module 3 · M3 · Q10–Q18 · 7 marks]

What is regression? Explain the principle of least squares, its assumptions, steps and types; compare with correlation.

Regression models the dependent variable Y as a function of independent variable(s) X to predict Y: $\hat{Y} = a + bX$. *Uses:* forecasting sales, price, risk, demand.

Least-squares principle. Choose a, b that minimise $SSE = \Sigma(y_i - a - bx_i)^2$. Setting $\partial SSE/\partial a = \partial SSE/\partial b = 0$ gives the **normal equations**: $\Sigma y = na + b\Sigma x$ and $\Sigma xy = a\Sigma x + b\Sigma x^2$.

Solution. $b = [n\Sigma xy - \Sigma x\Sigma y]/[n\Sigma x^2 - (\Sigma x)^2] = Sxy/Sxx$; $a = \bar{y} - b \cdot \bar{x}$.

Steps. (1) Tabulate x, y, xy, x². (2) Sum columns. (3) Solve normal equations for b then a. (4) Write $\hat{Y} = a + bX$. (5) Predict and check residuals / R^2.

Assumptions. Linearity; independence of errors; homoscedasticity (constant error variance); normally distributed errors; no perfect multicollinearity (multiple regression).

Types. Simple (1 predictor), multiple (many predictors), polynomial, logistic (classification), ridge/lasso (regularised).

Correlation vs regression. Correlation: symmetric, measures strength/direction, no dependent variable. Regression: asymmetric, gives an equation to predict Y from X, shows how much Y changes per unit X.

Answer: Minimise $\Sigma(y-a-bx)^2 \Rightarrow b = Sxy/Sxx$, $a = \bar{y} - b\,\bar{x}$

8. Regression numericals – Q15, Q19, Q20 [Module 3 · M3 · Q15, Q19, Q20 · 7 marks]

Fit Y = a + bX: (i) X=[1,2,3], Y=[2,3,5]; (ii) hours studied vs marks; (iii) price vs units sold, predict at ₹15.

(i) X=[1,2,3], Y=[2,3,5]. n=3, Σx=6, Σy=10, Σxy=2+6+15=23, Σx²=14. $b = (3 \cdot 23 - 6 \cdot 10)/(3 \cdot 14 - 36) = 9/6 =$ **1.5**; $a = 10/3 - 1.5 \cdot 2 =$ **0.333**. $\hat{Y} = 0.333 + 1.5X$.

(ii) Hours x=[1..5], marks y=[50,55,65,70,75]. n=5, Σx=15, Σy=315, Σxy = 50+110+195+280+375 = 1010, Σx² = 55. $b = (5 \cdot 1010 - 15 \cdot 315)/(5 \cdot 55 - 225) = (5050-4725)/50 =$ **6.5**; $a = 63 - 6.5 \cdot 3 =$ **43.5**. $\hat{Y} = 43.5 + 6.5X$ (each extra hour ≈ +6.5 marks).

(iii) Price x=[10,12,14,16,18], units y=[100,90,80,70,60]. n=5, Σx=70, Σy=400, Σxy = 1000+1080+1120+1120+1080 = 5400, Σx² = 100+144+196+256+324 = 1020. $b = (5 \cdot 5400 - 70 \cdot 400)/(5 \cdot 1020 - 4900) = (27000-28000)/200 =$ **−5**; $a = 80 - (-5)(14) =$ **150**.

Model: Units = 150 − 5·Price. **At ₹15:** 150 − 75 = **75 units**. Because the points lie exactly on a line, r = −1 and $R^2 = 1$.

Answer: (i) $\hat{Y}$=0.333+1.5X (ii) $\hat{Y}$=43.5+6.5X (iii) $\hat{Y}$=150−5X → 75 units at ₹15

9. Descriptive statistics numericals [Module 3 · M3 · Q1–Q13 · 7 marks]

Mean/median/mode, skewness from mean–median–mode, variance & SD, and effect of outliers.

Q7. Data 10,15,20,25,30. Mean = 100/5 = **20**; median (middle) = **20**; mode = **none** (all values occur once).

Q10. Mean 60 > Median 55 > Mode 50 → positive (right) skew. The tail stretches right, so a few large values pull the mean up (empirically mode ≈ 3·median − 2·mean = 45, close to 50).

Q12. Data 5, 8, 6, 10, 11. Range = 11−5 = **6**. Mean = 40/5 = 8. Deviations: −3, 0, −2, 2, 3 → squares 9, 0, 4, 4, 9 = 26.

Population variance = 26/5 = **5.2**, σ = **2.28**; sample variance = 26/4 = **6.5**, s = **2.55**.

Outliers (Q5, Q8). The mean is most affected (every value enters the sum); the median shifts only slightly; the mode is unaffected. e.g. adding 1000 to {10…30} lifts the mean from 20 to 183.3 but the median only to 22.5.

Kurtosis (Q11, Q13). High kurtosis ⇒ heavy tails and a sharper peak (more outliers than a normal curve). **Q15.** Sample 3,5,7,9,11 → point estimate of μ is $\bar{x}$ = 35/5 = **7**.

Answer: Q7: 20, 20, no mode · Q10: positive skew · Q12: range 6, var 5.2 (6.5), SD 2.28 (2.55) · Q15: 7

10. Maximum Likelihood Estimation – concept and steps [Module 3 · M3 · Q14–Q22 · 7 marks]

Define MLE, state the steps, why log-likelihood, and compare with method of moments.

Definition. The MLE $\hat{\theta}$ is the parameter value that makes the observed data most probable: $\hat{\theta} = \text{argmax } L(\theta)$ where $L(\theta) = \prod f(x_i; \theta)$ for i.i.d. data.

Steps. (1) Choose the model f(x;θ). (2) Write $L(\theta) = \prod f(x_i;\theta)$. (3) Take ln L (log-likelihood). (4) Differentiate: $\partial \ln L/\partial\theta = 0$. (5) Solve for $\hat{\theta}$. (6) Check second derivative < 0 (maximum).

Why log? Products become sums (easier derivative), avoids numerical underflow, and ln is monotonic so the maximiser is unchanged.

MLE vs Method of Moments. MoM equates sample moments to population moments (simple, closed-form, can be inefficient/biased). MLE maximises likelihood: consistent, asymptotically efficient and normal, invariant under re-parameterisation, but may need numerical optimisation.

Prefer MLE when the distribution is known and you need efficient estimates, e.g. logistic regression / neural-network cross-entropy training in AI (minimising cross-entropy = maximising likelihood).

Answer: $\hat{\theta} = \text{argmax } \prod f(x_i;\theta) \rightarrow$ solve $d \ln L/d\theta = 0$

11. MLE of Bernoulli / coin data [Module 3 · M3 · Q3, Q25 · 7 marks]

Coin results (i) H,H,H,T and (ii) H,T,H,H,H,T,H,H. Write L(p), ln L(p), find $\hat{p}$ and interpret.

(i) H,H,H,T: 3 heads, 1 tail. $L(p) = p^3(1-p)$. $\ln L = 3 \ln p + \ln(1-p)$. d/dp: $3/p - 1/(1-p) = 0 \Rightarrow 3(1-p) = p \Rightarrow$ **$\hat{p} = 3/4 = 0.75$**.

(ii) 8 flips: H,T,H,H,H,T,H,H → 6 heads, 2 tails.

(a) $L(p) = p^6(1-p)^2$. **(b)** $\ln L(p) = 6 \ln p + 2 \ln(1-p)$.

(c) $d \ln L/dp = 6/p - 2/(1-p) = 0 \Rightarrow 6(1-p) = 2p \Rightarrow 6 = 8p \Rightarrow$ **$\hat{p} = 0.75$**. Second derivative $-6/p^2 - 2/(1-p)^2 < 0 \Rightarrow$ maximum ✓.

(d) Interpretation. The MLE of P(heads) is simply the sample proportion of heads, h/n. At p = 0.75 the observed sequence is most likely: $L(0.75) = 0.75^6 \cdot 0.25^2 = 0.0111$, compared with L(0.5) = 0.0039.

Answer: $\hat{p}$ = heads / flips = 0.75 in both cases

12. MLE – Normal, Poisson, Binomial, Geometric [Module 3 · M3 · Q20, Q23, Q24, Q26, Q27 · 7 marks]

Find MLEs for μ (normal), λ (Poisson, prove), p (binomial 6/20), p (geometric [3,1,2,2,4]).

Normal mean (Q20), data 5,7,8,10,12. ln L = const − Σ(xᵢ−μ)²/2σ². d/dμ = Σ(xᵢ−μ)/σ² = 0 ⇒ $\hat{\mu} = \bar{x}$ = 42/5 = **8.4**.

Poisson proof (Q24). L(λ) = Π e^{−λ}λ^{xᵢ}/xᵢ!. ln L = −nλ + (Σxᵢ) ln λ − Σ ln xᵢ!. d/dλ = −n + Σxᵢ/λ = 0 ⇒ **$\hat{\lambda} = \Sigma x_i/n = \bar{x}$**. Second derivative −Σxᵢ/λ² < 0 ✓.

Binomial (Q26): n=20, x=6 defective. L(p) = C(20,6) p⁶(1−p)¹⁴. ln L = const + 6 ln p + 14 ln(1−p). 6/p = 14/(1−p) ⇒ $\hat{p}$ = 6/20 = **0.30**.

Geometric (Q23, Q27): P(X=x) = (1−p)^{x−1}p. For [3,1,2,2,4]: n = 5, Σx = 12. L(p) = pⁿ(1−p)^{Σx−n} = p⁵(1−p)⁷. ln L = 5 ln p + 7 ln(1−p). 5/p = 7/(1−p) ⇒ $\hat{p}$ = n/Σx = 5/12 = **0.4167**.

Answer: $\hat{\mu}$ = 8.4 · $\hat{\lambda} = \bar{x}$ · $\hat{p}$ = 0.30 (binomial) · $\hat{p}$ = 5/12 = 0.417 (geometric)

13. Bayes' theorem – total probability (3 sources) [Module 3 · M3 · Q11, Q12, Q13 · 7 marks]

Printers P1,P2,P3 (30/45/25 %; errors 1/2/4 %) – find P(P3 | misprint). Machines X,Y,Z (25/50/25 %; defects 1/2/6 %) – P(Z | defective). Suppliers A,B,C (40/35/25 %; defects 3/2/4 %) – P(A | defective).

Bayes: P(Aᵢ|E) = P(E|Aᵢ)P(Aᵢ) / Σ P(E|Aⱼ)P(Aⱼ).

Q11. Joint: P1: 0.30×0.01 = 0.003; P2: 0.45×0.02 = 0.009; P3: 0.25×0.04 = 0.010. P(misprint) = 0.022. P(P3 | misprint) = 0.010/0.022 = **0.4545 (45.5 %)**.

Q12. X: 0.25×0.01 = 0.0025; Y: 0.50×0.02 = 0.0100; Z: 0.25×0.06 = 0.0150. Total = 0.0275. P(Z | defective) = 0.0150/0.0275 = **0.5455 (54.5 %)**.

Q13. A: 0.40×0.03 = 0.012; B: 0.35×0.02 = 0.007; C: 0.25×0.04 = 0.010. Total = 0.029. P(A | defective) = 0.012/0.029 = **0.4138 (41.4 %)**.

Insight. The machine with the highest defect rate (or the largest volume × rate) dominates the posterior, even if it makes only a quarter of the output.

Answer: Q11: 0.4545 · Q12: 0.5455 · Q13: 0.4138

14. Bayes' theorem – tests and classifiers (base-rate effect) [Module 3 · M3 · Q6, Q8, Q14, Q17, Q18 · 7 marks]

Factory test, rare-disease test, COVID test and spam filter: compute the posterior probability.

Q14 – Factory. P(D)=0.30, P(+|D)=0.90, P(+|¬D)=0.05. P(+) = 0.30·0.90 + 0.70·0.05 = 0.27 + 0.035 = 0.305. P(D|+) = 0.27/0.305 = **0.885**.

Q17 – Rare disease (1 in 1000). P(+) = 0.001·0.99 + 0.999·0.02 = 0.00099 + 0.01998 = 0.02097. P(D|+) = 0.00099/0.02097 = **0.0472 (4.7 %)** – despite a “99 % accurate” test, a positive result still means only a ~5 % chance of disease, because the disease is so rare (base-rate fallacy).

Q18 – Spam filter. P(S)=0.20, P(flag|S)=0.90, P(flag|¬S)=0.05. P(flag) = 0.18 + 0.04 = 0.22. P(S|flag) = 0.18/0.22 = **0.818 (81.8 %)**.

Q8 – COVID. Prevalence 1 %, sensitivity 98 %, false-positive 2 %. P(+) = 0.01·0.98 + 0.99·0.02 = 0.0098 + 0.0198 = 0.0296. P(COVID|+) = 0.0098/0.0296 = **0.331 (33.1 %)**.

Q6 – Medical test 95 % accurate, 0.5 % prevalence (taking sensitivity = specificity = 0.95). P(+) = 0.005·0.95 + 0.995·0.05 = 0.00475 + 0.04975 = 0.0545. P(D|+) = 0.00475/0.0545 = **0.0872 (8.7 %)**.

AI use. Bayes underpins Naïve Bayes spam/text classifiers, medical decision support and Bayesian updating of model beliefs.

Answer: Q14: 0.885 · Q17: 0.047 · Q18: 0.818 · Q8: 0.331 · Q6: 0.087

15. Conditional, multiplication and addition rules [Module 3 · M3 · Q3, Q4, Q9, Q15, Q16 · 7 marks]

Solve: two red cards without replacement; P(English | Math); addition rule; both balls red.

Addition rule. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. **Multiplication rule.** $P(A \cap B) = P(A) \cdot P(B|A)$. Conditional: $P(B|A) = P(A \cap B)/P(A)$.

Q15 – two red cards, no replacement. $P(A) = 26/52 = 1/2$; $P(B|A) = 25/51$. $P(A \cap B) = (1/2)(25/51) = 25/102 =$ **0.2451**.

Q16 – P(English | Math). $P(M)=0.60$, $P(E)=0.50$, $P(M \cap E)=0.30$. $P(E|M) = 0.30/0.60 =$ **0.50**. Since $P(E|M) = P(E)$, passing English and Math are independent events.

Addition-rule example. $P(A)=0.3$, $P(B)=0.6$, $P(A \cap B)=0.2 \Rightarrow P(A \cup B) = 0.3 + 0.6 - 0.2 =$ **0.7**.

Balls (3 red, 2 green, as read from the page). $P(\text{both red}) = (3/5)(2/4) = 6/20 =$ **0.30**.

Answer: Q15: 0.2451 · Q16: 0.50 (independent) · $P(A \cup B) = 0.7$ · both red = 0.30

16. Supervised, unsupervised & reinforcement learning [Module 4 · M4 · Q5, Q8, Q25–Q27 · 7 marks]

Compare the three types of machine learning with examples and applications.

Machine learning: algorithms that learn patterns from data to predict or decide without being explicitly programmed.

Supervised. Labelled data (X, y); learns the mapping X→y. *Regression* (house price) and *classification* (spam / not spam, CAPTCHA digits). Algorithms: linear/logistic regression, k-NN, decision trees, SVM, Naïve Bayes.

Unsupervised. No labels; discovers structure. *Clustering* (k-means customer segmentation), *dimensionality reduction* (PCA), anomaly detection (fraud).

Reinforcement. An agent interacts with an environment, receives rewards and learns a policy that maximises cumulative reward. Examples: game-playing agents, robot navigation, recommendation tuning.

Semi-supervised. Few labels + many unlabelled points: cluster first, propagate labels, or use active learning (Fig 9.5: two labelled points → two clusters 'satisfied/not satisfied').

Application map. Prediction → regression; pattern detection → clustering/association; decision-making → reinforcement learning; automation & scalability → pipelines that retrain on new data.

Answer: Supervised = labelled · Unsupervised = unlabelled structure · RL = reward-driven policy

17. k-Nearest Neighbours – worked example, MSE, code [Module 4 · M4 · Q28–Q35 · 7 marks]

Explain k-NN, why it is lazy, role of Euclidean distance, effect of large k; classify (4 h study, 7 h sleep); compute MSE; Python snippet.

Principle. To classify a new point, compute its distance to every training point, take the k nearest, and vote (classification) or average (regression). **Lazy learner:** no training phase – it just stores data and does the work at prediction time.

Euclidean distance: $d = \sqrt{[(x_1-x_2)^2 + (y_1-y_2)^2]}$. (Scale features first!)

Q33 worked example (illustrative training set). Query = (4 h study, 7 h sleep). A(2,6,Fail): $d=\sqrt{5}=2.24$; B(3,7,Fail): $d=1.00$; C(5,6,Pass): $d=\sqrt{2}=1.41$; D(8,8,Pass): $d=4.12$; E(1,4,Fail): $d=4.24$. With $k=3$ the nearest are B, C, A → Fail, Pass, Fail ⇒ majority **Fail**. (Replace with your textbook table.)

Large k (Q32). $k=100$ on a small dataset → the vote covers almost the whole set, so predictions collapse to the majority class (under-fitting, high bias). Small k → noisy (over-fitting).

Q35 – MSE. $Y = [10,12,14]$, $\hat{Y} = [11,11,13]$. Errors = −1, +1, +1; squared = 1, 1, 1. $MSE = 3/3 =$ **1.0** (RMSE = 1.0).

Q31. Coefficient 2.3 for 'Advertising Spend': holding other variables constant, every 1-unit increase in ad spend raises the predicted outcome (e.g. sales) by 2.3 units.

```
from sklearn.neighbors import KNeighborsClassifier
X = [[2,6],[3,7],[5,6],[8,8],[1,4]]; y = ['Fail','Fail','Pass','Pass','Fail']
model = KNeighborsClassifier(n_neighbors=3).fit(X, y)
print(model.predict([[4,7]])) # ['Fail']
```

Q36. Test on unseen (hold-out) data to measure generalisation and detect over-fitting; **Q37.** prefer a simple model for interpretability, less data, speed and lower over-fitting risk.

Answer: Query (4,7) → Fail (k=3) · MSE = 1.0 · large k → majority class

18. Six steps of the data science process [Module 4 · M4 · Q1–Q11 · 7 marks]

Describe the six major steps of the data science process and why iteration matters.

1. Setting the research goal – understand the business context; formalise goals, deliverables and timeline in a **project charter**.

2. Retrieving data – find internal (databases, data warehouse) or external data; check data quality early.

3. Data preparation – *cleansing* (fix errors, missing values, duplicates), *integration* (joining, appending), *transformation* (log transform, dummy variables).

4. Data exploration (EDA) – histograms, boxplots, Pareto charts, correlation to find patterns and outliers before modelling.

5. Data modelling – select technique, execute (e.g. regression, k-NN), diagnose with R^2, adjusted R^2, p-values; main output = a trained, validated model.

6. Presentation & automation – communicate results with soft skills; automate scoring and reports.

Why iterate? Exploration reveals data problems and modelling reveals new questions, so earlier steps are revisited to improve data and model quality.

Answer: Goal → Retrieve → Prepare → Explore → Model → Present/Automate (iterative)

19. Big Data and the 5 Vs [Module 4 · M4 · Big Data Q1–Q30 · 7 marks]

Why do traditional tools struggle with Big Data? Explain the 5 Vs and the role of Value and Veracity.

5 Vs. Volume (terabytes–petabytes), **Velocity** (speed of arrival, real-time streams), **Variety** (structured, semi-structured, unstructured), **Veracity** (trustworthiness/quality) and **Value** (business insight extracted).

Why traditional tools fail. Single-server RDBMS cannot hold the volume (vertical scaling limit), cannot ingest fast streams, and require fixed schemas that do not fit text, images, audio or sensor data.

Solution. Horizontal scaling and distributed architecture (Hadoop/HDFS, Spark/PySpark), NoSQL stores, streaming tools (Kafka, Spark Streaming), batch + real-time processing together.

Why Value matters most. Data without insight is only cost; Value is the reason for collecting the other four Vs. **Veracity** – low-quality data (fake reviews, sensor noise) gives misleading analytics, critical in healthcare and finance.

Example – bank fraud detection: Velocity + Veracity most relevant; use Kafka + Spark Streaming + ML anomaly detection. **Hospital:** Variety – combine EHR tables (structured), lab reports (semi-structured), scans (unstructured) in a data lake.

Answer: Volume · Velocity · Variety · Veracity · Value